

7.11 PreQuiz: Logarithmic and Exponential Functions — Markscheme

1. [Maximum mark: 6]

Noah invests \$3200 at 7.2% per annum, compounded quarterly.

(a) Quarterly rate = $\frac{0.072}{4} = 0.018$. **M1**

$V(t) = 3200(1.018)^{4t}$ **A1**

(b) Since $(1.018)^{4t} = e^{4t \ln(1.018)}$, **M1**

$V(t) = 3200e^{kt}$ where $k = 4 \ln(1.018)$. **A1**

(c) Need the smallest integer n with $3200(1.018)^{4n} > 5000$. **M1**

$n > 6.25 \dots$, so the smallest full-year value is $n = 7$ years. **A1**

2. [Maximum mark: 6]

$$U(t) = 9000e^{kt} \text{ and } U(3) = 18\,500.$$

$$\textbf{(a)} \quad 9000e^{3k} = 18\,500, \text{ so } e^{3k} = \frac{18500}{9000}. \quad \textbf{M1}$$

$$k = \frac{1}{3} \ln\left(\frac{18500}{9000}\right) = 0.2401\dots \quad \textbf{A1}$$

$$\textbf{(b)} \quad U(7) = 9000e^{7k}, \text{ using their value of } k. \quad \textbf{M1}$$

$$U(7) = 48\,351.58\dots, \text{ so } 48\,352 \text{ users.} \quad \textbf{A1}$$

$$\textbf{(c)} \quad \text{Set } 9000e^{kt} = 50\,000. \quad \textbf{M1}$$

$$t = \frac{\ln(50000/9000)}{k} = 7.14 \text{ years.} \quad \textbf{A1}$$